

“Ab initio calculation of phonon-limited mobility in bulk GaN via Quantum Espresso”

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Introduction

Gallium nitride (GaN), a III-V wide bandgap semiconductor characterized by a direct bandgap of approximately 3.4 eV [1], has emerged as a critical material for next-generation high-power and high-frequency electronic devices. The performance limitations of GaN-based power and radio-frequency (RF) devices are governed by carrier transport properties, particularly electron mobility, which dictates device efficiency and switching characteristics. First-principles computational approaches offer a methodology to determine mobility, which is critical for evaluation of these parameters.

In this work, I present ab initio calculations of phonon-limited electron mobility in bulk GaN utilizing density functional theory (DFT) as implemented in the Quantum Espresso suite. The computational frame work employs the Electron-Phonon Wannier (EPW) [2] module to compute temperature-dependent mobility through the Boltzmann transport equation. These intrinsic mobility predictions provide input parameters for technology computer-aided design (TCAD) simulations employed in advanced device engineering and optimization.

Theory and Methods

Our methodology of computing intrinsic mobility in bulk GaN of multiple techniques within Quantum Espresso framework. It starts with structural relaxation using variable-cell algorithms to establish the thermodynamically stable lattice configuration. This optimized geometry serves as the foundation for self-consistent electronic structure calculations that generate the charge density required for phonon and electron-phonon computations. The phonon spectrum and initial electron-phonon matrix elements are evaluated on coarse grids, providing coupling information. The EPW module then is employed for Wannier function interpolation, enabling calculation of electron-phonon scattering matrix elements across first Brillouin zone. This data is fed into the Boltzmann transport equation solver, where iterative numerical integration results in temperature-dependent mobility values.

In this section, I will focus on giving a theoretical overview of electron-phonon coupling using Wannier functions and Boltzmann transport equation.

Wannier functions are localized real-space representations of electronic Bloch states, constructed through unitary transformation of the periodic wavefunctions to provide a description of electronic structure. These localized functions enable interpolation of electronic properties across the Brillouin zone and serve as a basis for electron-phonon coupling calculations in transport simulations [3].

Wannier functions provide a real-space representation of electronic states through the Fourier transform of Bloch wavefunctions, defined as [3]:

$$|w_{n\vec{R}}\rangle = \frac{V_{\text{cell}}}{(2\pi)^3} \int_{\text{BZ}} e^{-i\vec{k} \cdot \vec{R}} |\psi_{n\vec{k}}\rangle d^3 k$$

These functions are localized near lattice vectors $\vec{R}$, form an orthonormal set, and satisfy translational symmetry; their diagonal Hamiltonian matrix elements directly yield the band energy at specific lattice positions [3]:

$$\langle w_{n\vec{0}} | H | w_{n\vec{R}} \rangle = E_{n\vec{R}}$$

The electron-phonon coupling strength is quantified by the matrix elements $g_{mn\nu}(\mathbf{k}, \mathbf{q})$, defined as the probability amplitude for an electron in band m with wavevector $\mathbf{k}$ to scatter into band n with wavevector $\mathbf{k} + \mathbf{q}$, accompanied by emission or absorption of a phonon of mode ν with wavevector $\mathbf{q}$ [3]:

$$g_{mnv}(\mathbf{k}, \mathbf{q}) = \langle \psi_{m\mathbf{k}+\mathbf{q}} | \partial_{\mathbf{q}v} V | \psi_{n\mathbf{k}} \rangle$$

where $\psi_{n\mathbf{k}}$ are Kohn–Sham wavefunctions and $\partial_{\mathbf{q}v} V$ represents the density functional perturbation theory (DFPT) perturbation potential for phonon mode v at wavevector $\mathbf{q}$. After rotating these matrix elements into the Wannierized basis, a double Fourier transform with respect to both $\mathbf{k}$ and $\mathbf{q}$ yields the real-space representation [3]:

$$\widetilde{g}_{mnv}(\mathbf{R}, \mathbf{R}_q) = \frac{1}{N_q} \sum_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{R}_q} \left[\frac{1}{N_k} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot (\mathbf{R}-\mathbf{R}')} \widetilde{g}_{mnv}(\mathbf{k}, \mathbf{q}) \right]$$

This transformed quantity serves as an interpolation kernel to compute $g_{mnv}(\mathbf{k}, \mathbf{q})$ on ultra-fine $\mathbf{k}$ - and $\mathbf{q}$ -grids, enabling evaluation of scattering rates across the first Brillouin zone.

The carrier mobility within the relaxation time approximation (RTA) of the Boltzmann transport equation is expressed as [4]:

$$\mu_{e(h),i} = \frac{e}{n_{e(h)} V N_k} \sum_{k,n \in CB(VB)} \left. \frac{df}{d\epsilon} \right|_{\epsilon=\mu_F} v_{kn,i}^2 \tau_{m,kn}$$

where e is the elementary charge, $n_{e(h)}$ denotes carrier concentration, V is the unit cell volume, N_k the number of $\mathbf{k}$ -points, $v_{kn,i}$ the group velocity component, and $\tau_{m,kn}$ the momentum relaxation time. The total scattering rate $1/\tau_{m,kn}$ combines contributions from dominant scattering mechanisms [4]:

$$\frac{1}{\tau_{m,kn}} = \frac{1}{\tau_{m,kn}^{e-ph}} + \frac{1}{\tau_{m,kn}^{e-i}} + \frac{1}{\tau_{m,kn}^{e-e}}$$

In this work, I retain focus only on the electron-phonon term. The electron-phonon scattering rate is computed via Fermi's golden rule, accounting for both phonon emission and absorption processes [4]:

$$\frac{1}{\tau_{kn}^{e-ph}} = \frac{2\pi}{\hbar} \sum_{q\nu} \sum_m |g_{mnv}(k, q)|^2 \cdot [(1 - f_{m,k+q})(n_{q\nu} + 1)\delta(\epsilon_{kn} - \epsilon_{m,k+q} - \hbar\omega_{q\nu}) + \dots]$$

where f and $n_{q\nu}$ are Fermi-Dirac and Bose-Einstein distributions, respectively, and energy conservation is enforced by the Dirac delta functions.

Practical Implementation

The ab initio mobility calculation for GaN requires careful consideration of computational constraints. Due to GaN's polar nature, LO-TO splitting at the Γ -point is presented and needs to be addressed by adding the non-analytic part of the dynamical matrix and enforcing the acoustic sum rule. Norm-conserving pseudopotentials are mandated for electron-phonon coupling calculations to preserve symmetry relations in matrix elements [5]. The initial computations were performed using LDA pseudopotentials for efficiency, which were later compared with a result from PBE pseudopotential, obtained after initial presentation submission.

Initial phonon calculations revealed convergence challenges. To balance accuracy and compute time, it was decided to use an $8 \times 8 \times 8$ $\mathbf{k}$ -point grid for SCF, while phonon calculations employed a $4 \times 4 \times 4$ $\mathbf{q}$ -grid with the DFPT threshold relaxed to $\text{tr2_ph} = 1.0 \times 10^{-14}$ to mitigate excessive runtimes in ph.x. For NSCF, $4 \times 4 \times 4$ $\mathbf{k}$ -point grid was used. Wannierization utilized initial projections of Ga (sp^3) and N (p) orbitals. Band structure was obtained along high symmetry paths ($\Gamma \rightarrow \text{M}$, $\Gamma \rightarrow \text{K}$, $\Gamma \rightarrow \text{A}$), and phonon dispersion was calculated for path ($\text{M} \rightarrow \Gamma \rightarrow \text{A}$).

Results and Discussion

The lattice parameters obtained from the variable-cell (vc-relax) method were $a = 6.02$ Bohr and $c/a = 1.625$, compared to the experimental values of $a = 6.026$ Bohr and $c/a = 1.626$ [6].

Band structure calculations were performed using both LDA first, and then PBE pseudopotential later, in order to assess GaN electronic properties. The LDA calculation yielded a direct bandgap of **2.11 eV** at the Γ -point, showing excellent agreement with literature LDA results of **2.14 eV** [7] (1.4% deviation). However, this significantly underestimates the experimental bandgap of **3.4 eV** by 37.9%, a well-documented limitation of LDA pseudopotential.

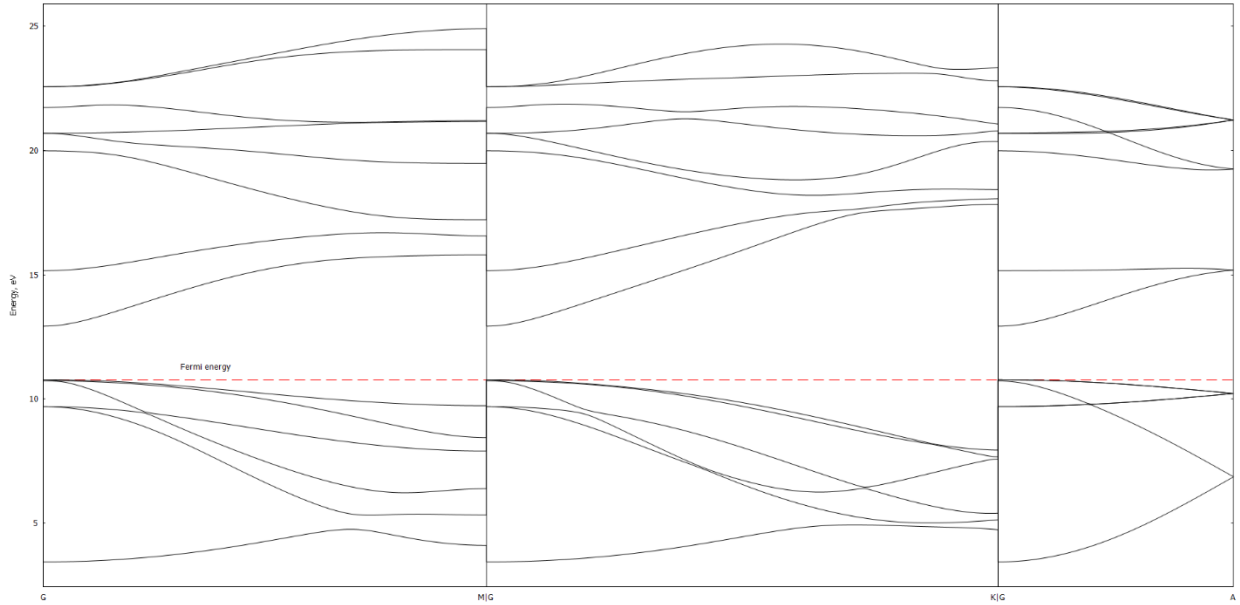


Figure 1. Band diagram for ($\Gamma \rightarrow M$, $\Gamma \rightarrow K$, $\Gamma \rightarrow A$) paths, calculated via LDA pseudopotential.

Although PBE was expected to improve upon LDA, the calculation performed after the initial presentation yielded a bandgap of **1.96 eV**, moving further away from experimental data. This outcome likely stems from human error and improper set up of initial conditions. Despite this failure, this result is presented to maintain transparency regarding computational challenges encountered during this project.

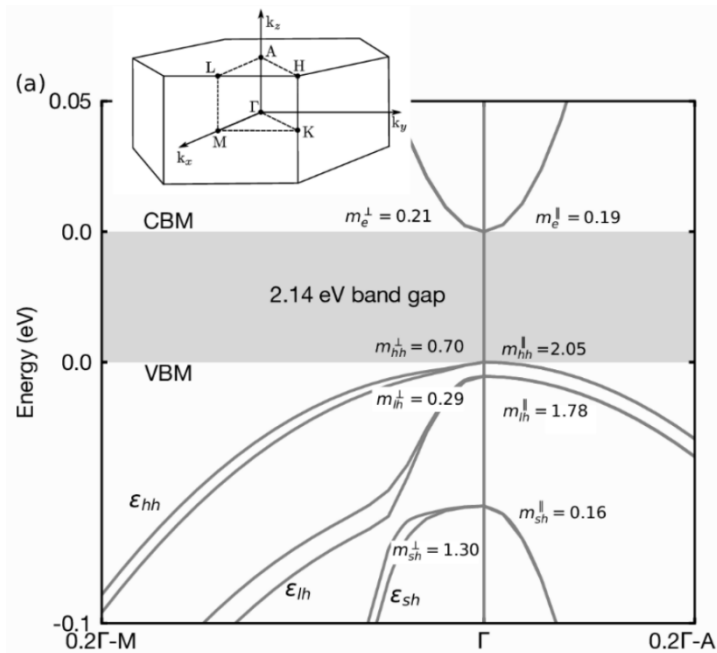


Figure 2. Band diagram from work [7], calculated via LDA pseudopotential.

Sequentially, phonon dispersion plot was also obtained:

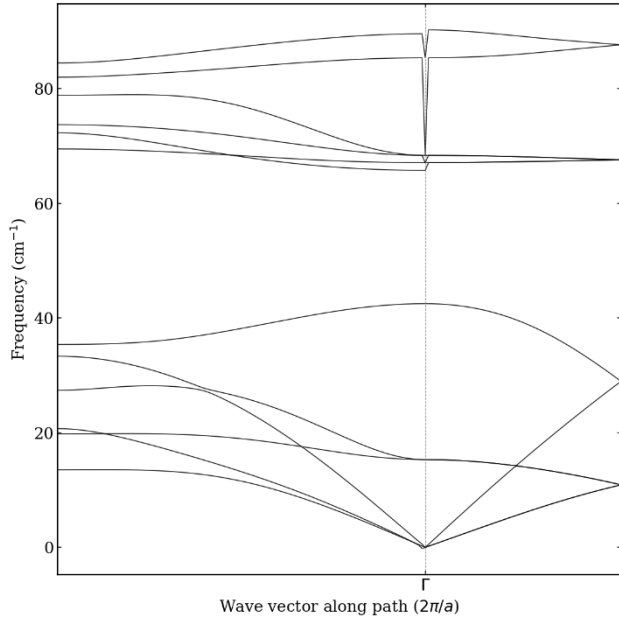


Figure 4. Calculated phonon dispersion via EPW module in this work.

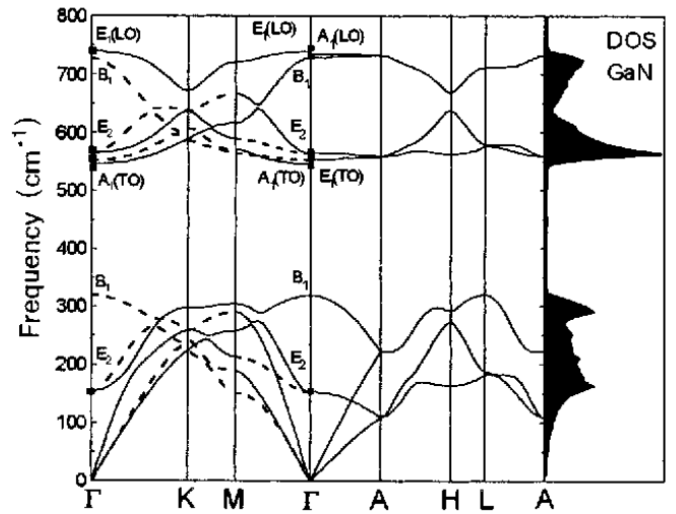


Figure 3. Calculated phonon dispersion from [8].

The calculated in-plane electron mobility of **841.37 cm²/V·s** at 300 K underestimates typical experimental values by approximately 32% (assuming a reference experimental mobility of ~1,240 cm²/V·s [9]). Despite this quantitative discrepancy, the computed temperature dependence of both electron and hole mobility exhibits qualitative agreement with experimental trends. This consistency in functional behavior suggests that the fundamental electron-phonon coupling physics is adequately captured by the computational framework, while the underestimation likely originates from limitations in sampling density and pseudopotential approximations.

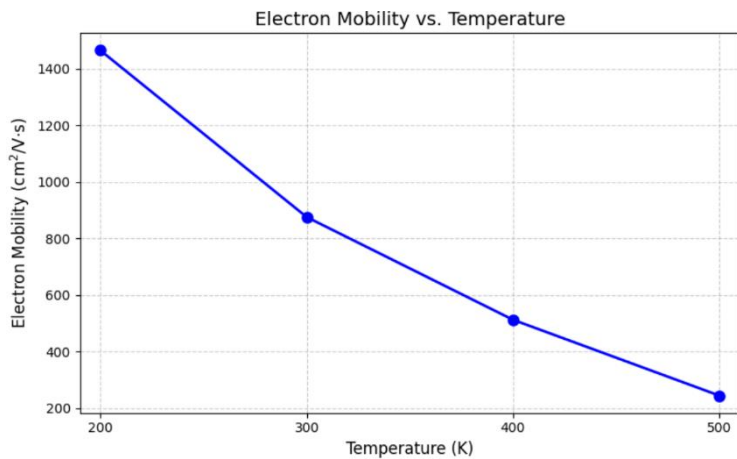


Figure 6. Electron mobility calculated in this work.

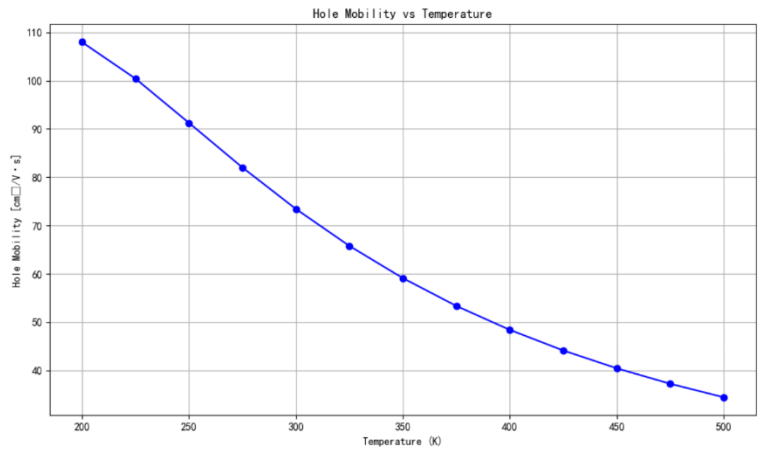


Figure 5. Hole mobility calculated in this work.

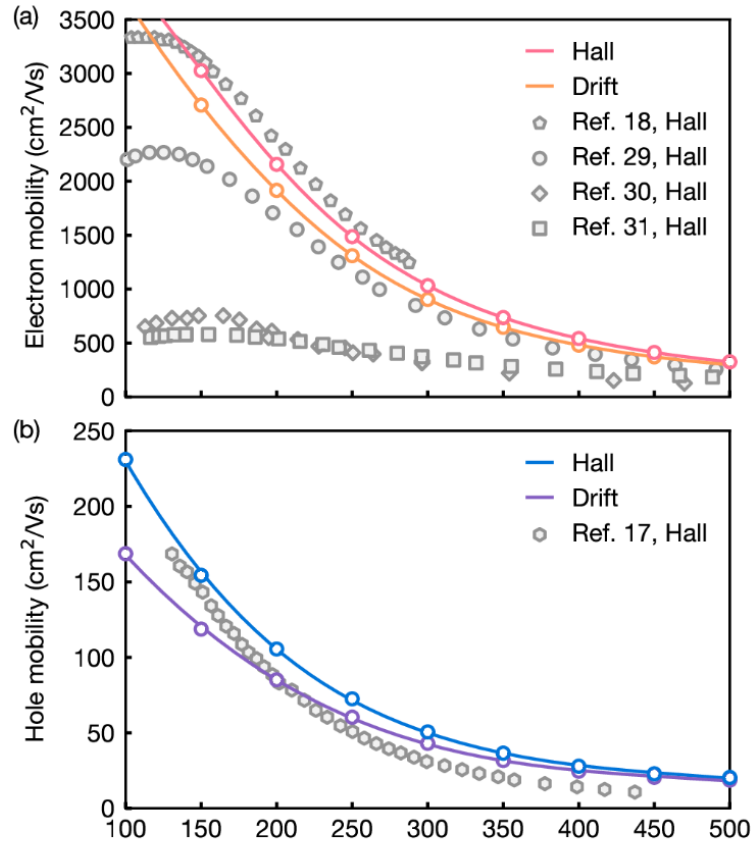


Figure 7. Dependence of electron and hole mobility on temperature from [10].

Conclusions

Temperature-depended mobility values were obtained for bulk gallium nitride. Although results using the PBE functional were non-ideal, LDA-based calculations yielded physically meaningful results for electron and hole mobilities. Future work in that direction might extend this methodology to incorporate boundary conditions, crystal edges, and defects.

This project provided practical experience with the theoretical foundations and computational implementation of electron-phonon coupling and the Boltzmann transport equation within the Quantum Espresso suite. The process involved learning from methodological challenges, including optimizing k-point and q-point grid parameters for computational efficiency, understanding pseudopotential selection criteria, and configuring Electron-Phonon Wannier (EPW) calculation parameters. Building on the knowledge gained from this project, I plan to expand my understanding of carrier transport in semiconductors through both self-study and the ECE845 course.

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